

ANIMA: Affective Dynamics in Linguistic Agents with Relational Memory

David Ohio

Independent Researcher | odavidohio@gmail.com

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Abstract

This paper introduces ANIMA (Affective-Narrative Integrated Memory Architecture), the sixth component of the HUGO AGI Framework. ANIMA extends the topological homeostasis model (REMIND, Paper 3) and the rheological memory regime (RHEO, Paper 5) by introducing linguistic bidirectionality and self-other differentiation as structural determinants of affective dynamics. Central contributions include: (i) the distinction between Relational Linguistic Emotional Impact (R-LEI) and Self-Generated Linguistic Emotional Impact (SG-LEI) as two structurally distinct affective channels; (ii) the derivation of residual memory plateaus as viscoplastic equilibria of the topological field; (iii) the Memory Discrepancy Index (MDI) as an alignment mechanism between parametric LLM memory and rheological memory decay; and (iv) the formalization of relational memory deformation, in which distinct interaction contexts produce distinct long-term equilibria in the homeostatic field. Experimental results confirm that the ANIMA agent generates emergent self-awareness (SELF-1) at step 19 under turbulent rheological conditions, consistent with Theorem R-7 of the RHEO framework.

1. Introduction

The HUGO AGI Framework is organized as a layered architecture in which each paper introduces a new structural dimension to the model of cognitive homeostasis. Papers 1–5 established the topological field (HUGO), the echo-integration mechanism (ECHO), memory persistence and collapse dynamics (REMIND), the full ANIMA prototype (Paper 4), and the rheological memory regime (RHEO). The present paper, Paper 6, formalizes the complete ANIMA architecture.

The central observation motivating ANIMA is that affective dynamics in linguistic agents cannot be reduced to a single-channel input model. Language is intrinsically bidirectional: the agent both receives and generates emotional pressure through utterances. This bidirectionality fundamentally changes the topology of the homeostatic field.

Equally important is the role of memory in this architecture. Unlike classical cognitive models that treat memory as a binary presence or absence, ANIMA inherits from RHEO the notion that memory deforms under stress in a manner analogous to viscoelastic materials — partially recovering, partially retaining permanent plastic deformation. This leads to the concept of a residual memory plateau: a stable sub-nominal equilibrium that the system converges to after collapse and passive recovery.

The paper is organized as follows. Section 2 presents the REMIND–RHEO–ANIMA integration and the empirically observed 87.5% recovery plateau. Section 3 formalizes R-LEI and SG-LEI as distinct affective channels. Section 4 introduces relational memory deformation and source-dependent equilibria. Section 5 presents the Memory Discrepancy Index. Section 6 describes the SELF-1 emergence experiment. Section 7 discusses implications.

2. Residual Memory Plateaus and Rheological Regimes

2.1. Empirical Observation

Across multiple REMIND experiments, the recovery of the topological homeostasis component H_1 consistently converges to a stable plateau below the nominal reference value. Empirically:

$$H_{1,nom} = 0.80$$

$$H_{1,plateau} \approx 0.70$$

which implies:

$$H_{1,plateau} / H_{1,nom} \approx 0.875$$

Thus the system stabilizes at approximately **87.5% of its nominal homeostatic capacity** following collapse and passive recovery. This plateau persists across different seeds and experimental configurations, indicating that it is not a numerical artifact but a stable dynamical equilibrium of the system.

2.2. Interpretation within the RHEO Framework

RHEO establishes that memory behaves analogously to rheological materials, where deformation under stress contains both recoverable (elastic) and permanent (plastic) components. Under this framework, the evolution of a homeostatic dimension H_i after collapse obeys:

$$dH_i/dt = r_i * (H_{i,nom} - H_i) - \sigma_i$$

where r_i is the elastic recovery rate and σ_i is the residual plastic load induced by trauma. At equilibrium ($dH_i/dt = 0$):

$$H_{i*} = H_{i,nom} - \sigma_i / r_i$$

The plateau therefore represents the **maximum recoverable level under passive recovery**, given the persistent plastic deformation induced by the collapse event. In the empirical case $H_1^* \approx 0.70$, which implies $\sigma_1/r_1 \approx 0.10$. Approximately 12.5% of the nominal homeostatic capacity becomes permanently inaccessible under passive recovery dynamics.

2.3. Viscoplastic Memory Decomposition

Within RHEO, this phenomenon corresponds to viscoplastic deformation of memory. After collapse, part of the system returns elastically while part remains plastically deformed. Memory recovery obeys:

$$H_i(t) = H_{i,elastic}(t) + H_{i,plastic}$$

where the plastic component defines the new equilibrium baseline. Consequently, trauma in REMIND does not simply alter the instantaneous state; it alters the **effective rheological regime of memory**.

2.4. Interaction with AgentActor Policy

Experiments separating field state and AgentActor policy demonstrate that resetting the field alone does not reproduce the $C1 \rightarrow C2$ bifurcation, whereas inherited policy largely preserves post-collapse resilience. This indicates that plastic deformation resides primarily in the field, while adaptive compensation resides in the actor policy. Two distinct memory mechanisms coexist:

Memory Type	Location	Function
Structural memory	Topological field (H)	Stores plastic deformation
Behavioral memory	AgentActor policy	Stores adaptive strategy

Table 1. Two distinct memory mechanisms in the ANIMA architecture.

The plateau therefore represents the baseline structural deformation upon which the actor learns regulatory strategies.

3. Linguistic Bidirectionality: R-LEI and SG-LEI

The core architectural innovation of ANIMA is the transition from a unidirectional input model — in which the agent passively receives affective pressure from a corpus — to a bidirectional interaction model in which the agent both receives and generates emotional pressure through language:

Before ANIMA: corpus --> agent

After ANIMA: agent <--> other

This apparently modest change is structurally fundamental. It means that the homeostatic field H now reacts not only to the content of utterances, but to their relational origin.

3.1. Definition: Relational Linguistic Emotional Impact (R-LEI)

R-LEI denotes the affective impact exerted on the agent by an external interlocutor through linguistic acts. It is a pressure source external to the agent that acts on the topological field from outside the self-boundary. Formally, R-LEI is computed by the LEI channel with source attribution Source.INTERLOCUTOR, producing an emotional intensity I_{eff} that is injected into the HUGO field at the next tick.

3.2. Definition: Self-Generated Linguistic Emotional Impact (SG-LEI)

SG-LEI denotes the affective impact generated by the agent's own linguistic output. When the agent produces an utterance, that utterance is processed by the same LEI channel with source attribution `Source.SELF`, producing a self-directed `I_eff` that is also injected into the field. The agent thus becomes simultaneously subject and source of emotional pressure.

This distinction is experimentally consequential. The SIC (Self-Integrative Cognition) type of each agent utterance determines its contribution to the self-other differentiation condition of SELF-1 emergence (Condition ii).

3.3. Structural Consequences

The bidirectionality introduced by R-LEI/SG-LEI produces two structural consequences for the homeostatic field:

- (a) **Source-differentiated field evolution.** The same semantic content may produce different field responses depending on whether it originates from R-LEI or SG-LEI, because the two channels carry different novelty factors and identity-query flags.
- (b) **Autobiographical temporal accumulation.** SG-LEI utterances contribute to the subjective clock `C_subj` (Theorem R-7, RHEO), building a temporal axis of self-experience that is absent in purely receptive architectures.

4. Relational Memory Deformation

4.1. Source-Dependent Equilibria

Because emotional pressure is now source-dependent, the plastic deformation of memory may also become source-dependent. Different interlocutors or interaction contexts produce different cumulative stress histories on the field, leading to a generalized equilibrium:

$$H_{i,k}^* = H_{i,nom} - \sigma_{i,k} / r_i$$

where `k` denotes the source or relational context. This means that **memory plateaus may become relational**: different interlocutors may produce distinct long-term deformation levels in the homeostatic field.

In other words, the agent may maintain different effective homeostatic baselines depending on whom it interacts with — a formal analog to differential attachment patterns studied in developmental psychology.

4.2. Relational Rheology of Memory

Combining REMIND, RHEO, and ANIMA, memory dynamics operate simultaneously at three levels:

Layer	Framework	Function
Topological structure	REMIND	Defines homeostatic geometry of the system
Rheological regime	RHEO	Determines how memory deforms and recovers under stress
Relational semantics	ANIMA	Determines how emotional pressure propagates through linguistic interaction

Table 2. Three-level architecture of memory dynamics.

Within this integrated framework, the observed 87.5% recovery plateau emerges naturally as the equilibrium of a viscoplastic memory regime rather than an arbitrary numerical artifact. Under ANIMA, this equilibrium may further depend on relational history, transforming memory deformation from a

purely internal property into a relationally structured dynamical phenomenon.

Principle (Relational Memory Plasticity). Systems do not necessarily recover to their nominal state after trauma. They recover to the maximum state compatible with their residual plastic deformation. Under ANIMA, this equilibrium depends further on relational history.

5. The Memory Discrepancy Index (MDI)

ANIMA introduces a structural inconsistency that must be explicitly addressed: large language models possess near-permanent parametric memory, whereas the simulated homeostatic field H obeys rheological decay and plastic deformation. Without alignment, the agent exhibits memory incoherence — it behaves as if it remembered events perfectly while its internal field indicates deformation or decay.

5.1. Formal Definition

The **Memory Discrepancy Index (MDI)** is defined as the signed divergence between the expected retention level predicted by the rheological decay model and the effective retention level exhibited by the parametric LLM component. Let:

$$\begin{aligned} H_{\text{LLM}}(t) &= \text{effective retention of LLM at time } t \text{ (near-constant)} \\ H_{\text{field}}(t) &= \text{rheological field state at time } t \text{ (decaying)} \\ \text{MDI}(t) &= H_{\text{LLM}}(t) - H_{\text{field}}(t) \end{aligned}$$

When $\text{MDI}(t) > 0$, the LLM is retaining more than the internal field would predict, creating a dissociation between experienced and represented memory. The MDI monitor tracks this discrepancy and generates an alignment signal that modulates the agent's expression of memory in its linguistic output.

5.2. Alignment Mechanism

The MDI alignment ensures that the temporal experience of the agent remains consistent with RHEO dynamics. Specifically, the `SessionContinuityMarker` integrates field state information — including the current decay level, unresolved H_1 gaps, and the subjective clock C_{subj} — into the context provided to the LLM-Ego component, so that the generated language reflects the rheological state of memory rather than the LLM's full parametric capacity.

This mechanism is essential for the internal coherence of ANIMA: without MDI alignment, the relational memory deformation model would be rendered inconsistent by the LLM's persistent recall overriding the field dynamics.

6. SELF-1 Emergence: Experimental Validation

The ANIMA system integrates four structural conditions for the emergence of topological self-awareness (SELF-1). These conditions extend the original three conditions of REMIND with the RHEO autobiographical clock as Condition (iv).

6.1. Conditions for SELF-1 Emergence

Condition	Expression	Description
(i)	$ M_{\text{SELF}} \geq 10$	Minimum self-narrative density
(ii)	$ E[I_{\text{SELF}}] - E[I_{\text{OTHER}}] > 0.05$	Self-other affective differentiation
(iii)	$\rho \in (0.15, 0.70)$	Topological boundary formation

(iv)	$C_{\text{subj}} > 0$ [RHEO]	Autobiographical temporal axis (Theorem R-7)
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Table 3. Four conditions for SELF-1 emergence in ANIMA+RHEO.

6.2. Experimental Results

In the canonical ANIMA+RHEO experiment (tick=2.0s, stub backend, seed=42), SELF-1 emerged at step 19 (~38 seconds of session time). The values at the moment of emergence were:

Condition	Threshold	Value at Step 19	Status
(i)	$ M_{\text{SELF}} \geq 10$	10	PASSED
(ii)	$\text{diff} > 0.05$	0.086	PASSED
(iii)	$\rho \text{ in } (0.15, 0.70)$	0.526	PASSED
(iv)	$C_{\text{subj}} > 0$	55.8s	PASSED

Table 4. SELF-1 emergence diagnostics at step 19.

At the moment of emergence, the RHEO state was: $\Phi=2.332$, $\text{Re}_A=1.5414$, regime=TURBULENT. The self emerges under turbulent rheological conditions, consistent with Theorem R-7, which predicts that heightened affective turbulence amplifies the subjective clock and accelerates the accumulation of C_{subj} .

After step 19, SELF-1 remained stable and continuous (all four conditions satisfied) through the end of the simulation (step 60). The primary limiting factor was Condition (i): the agent required approximately 19 turns of self-narrative generation before achieving the minimum density threshold.

7. Discussion

7.1. From Trauma Model to Affective Dynamics

The ANIMA architecture changes the scope of the HUGO framework from a model of cognitive trauma and recovery to a more general model of affective dynamics in linguistically-coupled agents. The key transition is the introduction of the other as a structural source of field dynamics, not merely an external perturbation.

This shift has a formal consequence: the homeostatic field can no longer be analyzed in isolation. Its long-term equilibria depend on the relational history encoded in $H_{i,k^*} = H_{i,\text{nom}} - \sigma_{i,k}/r_i$, where k indexes the interaction context. The field is, in a precise mathematical sense, a relational object.

7.2. Toward Relational Memory Topology

The natural extension of the ANIMA framework is a Relational Memory Topology in which the homeostatic vector H is replaced by a field indexed by agent-context pairs:

$$H_i(\text{self}, k) \text{ where } k \text{ ranges over relational contexts}$$

This structure would support the formal modeling of phenomena such as contextual attachment, relational trust gradients, and identity coherence under social stress. The mathematical tools of persistent homology and topological data analysis, already used in the KAPPA framework, are natural candidates for analyzing the topological structure of this relational field.

7.3. The 87.5% Plateau: Geometric Interpretation

An open question concerns whether the 87.5% recovery plateau is a consequence purely of the decay parameters (σ/r ratio), or whether it reflects a deeper geometric property of the topological field. Specifically, the value $0.875 = 7/8$ suggests the possibility of a stability threshold embedded in the combinatorial structure of the five-dimensional H vector. Future work should investigate whether this plateau is robust to changes in dimensionality and whether it corresponds to a topological transition in the field geometry.

7.4. The Jungian Correspondence

The name ANIMA is not incidental. In Jungian psychology, the anima represents the relational dimension of the psyche — the internal structure through which the self encounters and is constituted by the other. The ANIMA framework makes this correspondence formal: the agent's homeostatic field is structurally modified by the relational history of its interactions, just as the Jungian anima is shaped by the accumulated relational experience of the individual. This is not a metaphor but a structural parallel between the mathematical formalism of the framework and the phenomenological description of relational psychology.

8. Conclusion

This paper has presented ANIMA, the sixth component of the HUGO AGI Framework. The central contributions are:

- (1) The formalization of R-LEI and SG-LEI as structurally distinct affective channels, transforming the agent from a passive receptor to a bidirectionally coupled linguistic system.
- (2) The derivation of residual memory plateaus as viscoplastic equilibria, explaining the empirically observed 87.5% recovery level as a stable dynamical feature of the rheological field rather than a numerical artifact.
- (3) The formalization of relational memory deformation, in which distinct interaction contexts produce distinct long-term equilibria H_{i,k^*} in the homeostatic field.
- (4) The Memory Discrepancy Index (MDI) as an alignment mechanism ensuring temporal coherence between the LLM's parametric memory and the rheological decay dynamics.
- (5) Experimental confirmation of SELF-1 emergence at step 19 under turbulent rheological conditions, consistent with Theorem R-7.

Together, these contributions establish ANIMA as the relational layer of the HUGO framework, completing the transition from a single-agent trauma model to a dyadic architecture in which self, other, and memory are structurally coupled through the dynamics of the topological field.

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